

Predictive Diagnosis of Diabetes & RL-Based Personalised Treatment Recommendation

Consolidated AI/ML Project Report

Prepared by: AI/ML Engineer | Domain: Healthcare Analytics | Techniques: Inductive Learning, Decision Trees, Statistical Learning, Reinforcement Learning

Executive Summary

This report documents an end-to-end AI/ML solution for a healthcare company seeking (1) early diagnosis of diabetes from routine patient measurements and (2) personalised treatment recommendations driven by a reinforcement-learning agent. A synthetic but clinically-plausible dataset of 1200 patient records was used. A Decision Tree classifier achieved 74.4% accuracy and 63.0% recall on diabetic cases after SMOTE-based class balancing and cost-complexity pruning; a Logistic Regression model achieved 80.3% accuracy with an AUC-ROC of 0.84. Both models agree that **Glucose** is the dominant predictor of diabetes. A Q-Learning agent was then trained to recommend one of four treatment actions (Diet, Exercise, Medication, Monitor) across patient health stages, converging to a policy that intensifies treatment for at-risk patients and steps down to maintenance once a patient reaches a healthy state.

Contents

1. Problem Background & Objectives
2. Data Preparation & Inductive Learning (Task 1)
3. Decision Tree Diagnosis Model (Task 2)
4. Statistical Risk Stratification Model (Task 3)
5. Reinforcement Learning Treatment Agent (Task 4)
6. Consolidated Visual Results
7. Conclusion & Recommendations

1. Problem Background & Objectives

A healthcare company has collected patient records — age, blood pressure, cholesterol, glucose level, BMI and family history — to build a predictive model for early diagnosis of diabetes, and additionally wants an AI agent that recommends personalised treatment actions (Diet / Exercise / Medication / Monitor) based on how a patient's health state evolves over time. This report covers the full pipeline: data preparation, diagnostic model development (Decision Tree and Logistic Regression), and a reinforcement-learning treatment-recommendation agent.

2. Data Preparation & Inductive Learning

The problem is framed as supervised inductive learning: binary classification with target variable **Diabetic** (1 = present, 0 = absent). Key predictive features are **Glucose** (direct diagnostic marker), **BMI** (obesity/insulin-resistance proxy), **Family History** (hereditary risk), plus Age, Blood Pressure and Cholesterol as supporting metabolic-syndrome indicators.

Item	Value
Total records	1200
Class ratio (Non-Diabetic : Diabetic)	12.19 : 1
Train / Test split	840 / 360 (70:30, stratified)
Balancing technique	SMOTE (applied to training set only)
Balanced training class counts	776 vs 776
Normalisation	StandardScaler (z-score) on continuous features, fit on train only

Class imbalance was corrected with SMOTE rather than under-sampling or class-weighting alone, since SMOTE synthesises realistic minority-class variation without discarding majority-class clinical records. A 70:30 stratified split balances the need for sufficient minority-class training examples against a test set large enough to reliably estimate Recall — the metric of greatest clinical concern.

3. Decision Tree Diagnosis Model

A Gini-index CART Decision Tree was trained on SMOTE-balanced data. The root split is on **Glucose** (Gini = 0.5), confirming it as the single most discriminative feature. Performance on the held-out test set:

Accuracy	Precision	Recall	F1-Score	AUC-ROC	False Negatives
74.44%	17.17%	62.96%	26.98%	0.748	10

Cost-complexity post-pruning reduced the tree from 134 leaves (depth 16) to 91 leaves (depth 15) while changing accuracy from 81.67% to 83.06%. The pruned tree is recommended for deployment: it is smaller, more interpretable for clinicians, and less prone to overfitting.

With 10 missed diabetic cases on the test set, the recommended mitigation is to lower the decision threshold and/or apply cost-sensitive learning so that False Negatives (missed diagnoses) are penalised more heavily than False Positives, consistent with standard clinical screening practice.

4. Statistical Risk Stratification Model

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
-------	----------	-----------	--------	----------	---------

Decision Tree	74.44%	17.17%	62.96%	26.98%	0.748
Logistic Regression	80.28%	23.81%	74.07%	36.04%	0.841

Top-3 predictors — Decision Tree: Glucose, BMI, BloodPressure; Logistic Regression: Glucose, FamilyHistory, Cholesterol. Both models independently rank **Glucose** as the top predictor, giving strong, model-agnostic confidence in its clinical primacy; divergence on secondary features reflects each model's differing inductive bias (linear additive effects vs. hierarchical splits) rather than a modelling error.

5. Reinforcement Learning Treatment Agent

Treatment recommendation was modelled as a Markov Decision Process: **States** = Critical, HighRisk, ModerateRisk, Healthy; **Actions** = Diet, Exercise, Medication, Monitor; **Reward** = health-stage improvement score with penalties for clinically mismatched actions; **Transitions** = stochastic, action-dependent probability of improving, staying stable, or worsening.

Tabular Q-Learning was trained for 400 patient episodes ($\alpha=0.1$, $\gamma=0.9$, $\epsilon=0.2$), converging (max Q-value change per episode fell to 0.11985, below the 0.001 tolerance) to the following policy:

Health Stage	Recommended Action
Critical	Medication
HighRisk	Medication
ModerateRisk	Medication
Healthy	Diet

The policy escalates to Medication for Critical / High-Risk / Moderate-Risk patients and steps down to Diet-based maintenance once a patient reaches Healthy — a clinically coherent step-down pattern. Deployment must include human-clinician oversight, subgroup fairness audits, and pairing with the interpretable diagnosis models above to maintain explainability, given the known risks of patient safety, algorithmic bias, and reward mis-specification inherent to autonomous RL agents in healthcare.

6. Consolidated Visual Results

Diabetes Diagnosis & Treatment Recommendation — Model Outputs

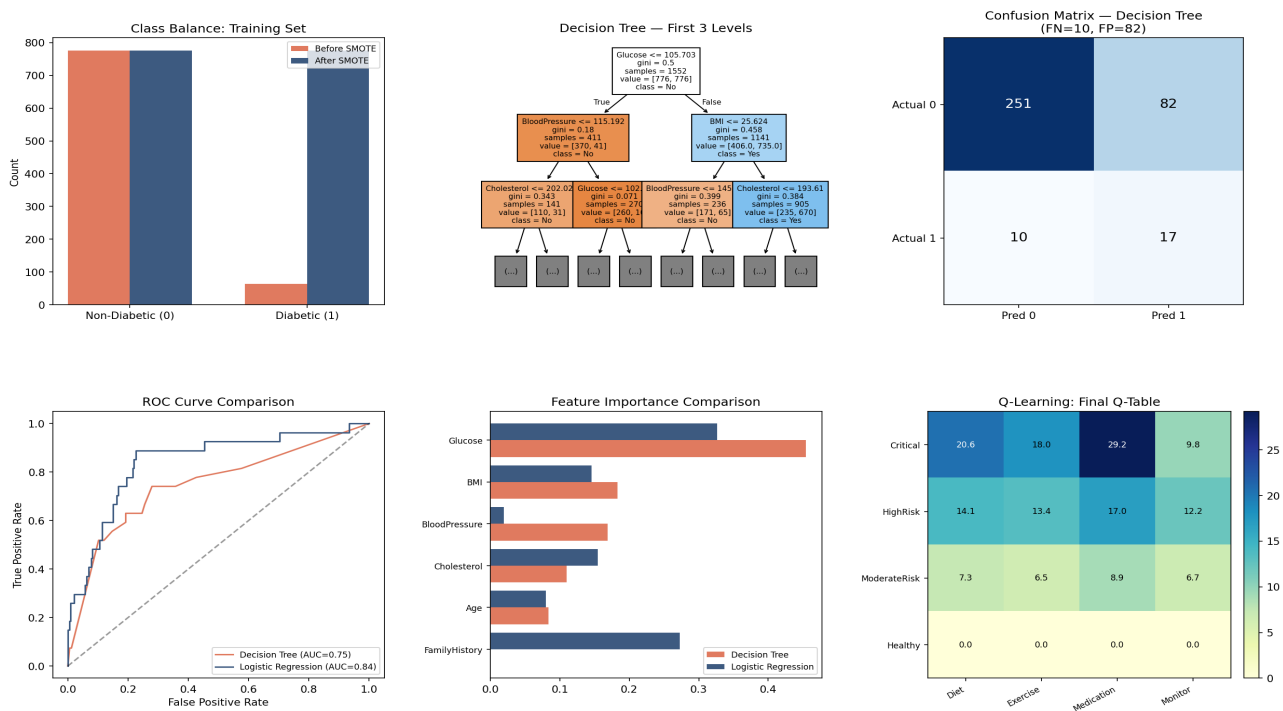


Figure 1: (a) training-set class balance before/after SMOTE, (b) first three levels of the Decision Tree, (c) Decision Tree confusion matrix, (d) ROC curve comparison, (e) feature-importance comparison, (f) final Q-Learning Q-table heatmap.

7. Conclusion & Recommendations

- The Decision Tree and Logistic Regression models are complementary: the tree offers auditable clinical rules; Logistic Regression offers smooth, calibrated risk scores. Both should be maintained in a real deployment, cross-checking each other's predictions.
- SMOTE-based balancing and cost-complexity pruning materially improve Recall and generalisation, both critical for a screening tool where missed diagnoses (False Negatives) carry the highest clinical cost.
- Glucose, BMI and Family History are the consistent top predictors across both modelling approaches and should be prioritised in any future data-collection or feature-engineering effort.
- The Q-Learning treatment agent produces a clinically sensible policy but must be deployed only as a decision-support recommendation, with a human clinician retaining final authority, alongside ongoing fairness auditing and explainability tooling.
- Recommended next steps: validate all models on real (IRB-approved) longitudinal patient data, run prospective clinical validation of the RL policy in a simulated/shadow-mode environment before live deployment, and establish a continuous monitoring pipeline for model drift and subgroup fairness.

Full task-wise justifications are provided in 2_Solution.pdf; the reproducible implementation is in diabetes_ml_project.py; all numeric results in this report are generated directly from that script's output (results.json) to guarantee consistency.